

PROJECT ANALYSIS REPORT

Bagisto

Laravel E-Commerce Framework

A comprehensive technical analysis of the Bagisto open-source Laravel e-commerce framework, covering architecture, modules, dependencies, codebase metrics, and deployment readiness assessment.

Repository: github.com/bagisto/bagisto

Latest Version: v2.4.6 (June 5, 2026)

License: MIT

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1. Executive Summary

Bagisto is a comprehensive, open-source Laravel-based e-commerce framework that provides a robust foundation for building online stores and marketplaces. The project leverages modern web technologies including Laravel 12, Vue.js, and Vite, and follows a modular architecture pattern with over 40 independent packages. This report presents a thorough analysis of the Bagisto codebase, covering its architecture, modules, dependencies, code quality metrics, and deployment readiness.

The framework is designed to serve a wide range of e-commerce needs, from single-vendor shops to multi-vendor marketplaces, B2B commerce platforms, and headless commerce solutions. It supports multiple payment gateways including Stripe, PayPal, Razorpay, PayU, and PhonePe, and includes advanced features such as booking products, RMA (Return Merchandise Authorization), GDPR compliance, and AI-powered content generation through its MagicAI module. The project is actively maintained with regular releases, the latest being v2.4.6 released on June 5, 2026.

40+ Modules	474K+ PHP LOC	2,434 PHP Files	551 Blade Templates	v2.4.6 Latest Version
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2. Project Overview

Bagisto positions itself as a developer-friendly, modular e-commerce framework built on top of the Laravel PHP framework. Unlike monolithic e-commerce platforms, Bagisto adopts a package-based architecture where each functional domain (products, orders, customers, payments, etc.) is implemented as an independent Laravel package under the Webkul namespace. This design choice enables developers to selectively enable or disable features, extend functionality through package overrides, and maintain a clean separation of concerns across the entire application.

The project is licensed under the MIT license, making it fully open-source and free for commercial use. It has a vibrant community with active contributions on GitHub, a dedicated forum, and a Facebook community group. The framework supports over 20 languages including Arabic, Chinese, English, French, German, Japanese, and Russian,

making it suitable for global e-commerce deployments. The official documentation is available at devdocs.bagisto.com, and a live demo is accessible at demo.bagisto.com for quick evaluation.

Key Project Information

Project Name	Bagisto
Repository	github.com/bagisto/bagisto
Description	Laravel E-Commerce Framework
License	MIT
Latest Version	v2.4.6 (June 5, 2026)
PHP Requirement	PHP 8.3+
Framework	Laravel 12
Frontend	Vue.js + Vite 6
Database	MySQL 8.0
Cache	Redis (Predis)
Search	Elasticsearch 7.17
Total Size	502 MB (with git history)
Supported Languages	20+ including Arabic, Chinese, English

3. Technical Architecture

Bagisto follows a modular monolith architecture, where the entire application is deployed as a single unit, but internally organized as a collection of independent packages. This approach combines the operational simplicity of a monolithic deployment with the code organization benefits of a modular system. Each package is self-contained with its own models, repositories, controllers, views, migrations, and service providers, following the Laravel convention of modular package development.

The architecture relies heavily on the Laravel service provider pattern for package registration and bootstrapping. Each module registers its own routes, views, configurations, and event listeners through dedicated service providers. The Konekt

Concord package is used as the module management layer, providing a standardized way to discover and register modules. Additionally, the Astrotoomic Laravel Translatable package enables multi-language support at the model level, allowing product descriptions, category names, and other translatable content to be stored and retrieved in multiple languages seamlessly.

Architecture Layers

Layer	Technology	Description
Frontend	Vue.js + Vite 6	Reactive UI components and asset bundling
Template Engine	Blade (551 files)	Server-side rendering for admin and shop views
Application Layer	Laravel 12	MVC framework, service providers, middleware
Module System	Konekt Concord	Package discovery and registration
Data Layer	Eloquent ORM	Active Record pattern with repository abstraction
Data Grid	Custom DataGrid	Server-side data tables with filtering and export
Cache Layer	Redis + FPC	Full Page Cache with event-based invalidation
Search Engine	Elasticsearch 7.17	Full-text search with Kibana dashboard
Queue System	Laravel Queue	Async job processing for imports, indexing, emails
Authentication	Sanctum + Socialite	Token-based auth and social login

Docker Infrastructure

Bagisto provides a comprehensive Docker Compose configuration for local development using Laravel Sail. The Docker setup includes the following services: a PHP 8.3 application server, MySQL 8.0 database, Redis cache, Elasticsearch 7.17 with Kibana for search and analytics, and Mailpit for email testing. This containerized environment ensures consistency across development machines and simplifies the onboarding process for new contributors. The Elasticsearch instance is configured with 256MB heap size and single-node discovery mode, suitable for development but requiring tuning for production deployments.

4. Module Analysis

The Bagisto framework is composed of 40 independent packages organized under the Webkul namespace. Each package follows a consistent internal structure with `src/`, `tests/`, and `resources/` directories. The packages are designed around domain-driven design principles, where each package encapsulates a specific business domain with its own data models, business logic, and presentation layer. This modular approach allows developers to customize or replace individual packages without affecting the rest of the system.

Core Modules

Module	Purpose	Key Features
Admin	Admin panel	Dashboard, CRUD operations, configuration management (553 PHP files)
Shop	Storefront	Customer-facing catalog, checkout, product pages (329 PHP files)
Core	Foundation	Base classes, helpers, channel management (171 PHP files)
Product	Product management	Simple, configurable, grouped, bundle, downloadable types
Category	Category management	Nested set tree (kalno/nestedset), translations
Customer	Customer accounts	Registration, auth, wishlist, compare, groups, VAT validation
Sales	Order management	Orders, invoices, shipments, refunds
Checkout	Cart & checkout	Multi-step checkout, address management, cart rules
Inventory	Stock management	Inventory sources, multi-warehouse tracking
Payment	Payment abstraction	Base payment interface, COD, money transfer

Payment Gateways

Bagisto integrates with multiple payment providers through a standardized payment interface. Each payment gateway is implemented as a separate package, allowing for easy addition or removal of payment methods. The payment packages include Stripe (using the official stripe-php SDK v17.3), PayPal (via PayPal Server SDK v2.0), Razorpay (for the Indian market), PayU (another Indian payment gateway), and PhonePe (UPI-based payments). All payment packages follow a consistent pattern with dedicated controllers, configuration files, and localized language files supporting over 15 languages each. The CashOnDelivery and MoneyTransfer methods are provided as built-in offline payment

options.

Marketing & SEO Modules

The framework includes comprehensive marketing and SEO tools. The CartRule module provides discount and coupon management with support for percentage and fixed-amount discounts, coupon code generation, and usage limits. The CatalogRule module enables automated price rules based on conditions such as customer groups, channels, and product attributes. The Marketing package consolidates promotional tools, while the Sitemap module generates XML sitemaps for search engine optimization. The SocialLogin package enables authentication via social media platforms, and SocialShare adds sharing buttons for products across social networks.

Advanced Features

Module	Description
MagicAI	AI-powered content generation supporting OpenAI, Gemini, Anthropic, Mistral, Ollama, Groq, DeepSeek, and xAI models
BookingProduct	Booking system with appointment, event, rental, default, and table slot types
RMA	Return Merchandise Authorization with custom fields, status tracking, and resolution management
GDPR	GDPR compliance with data request management and consent tracking
EUWithdrawal	EU withdrawal directive compliance (Directive 2023/2673)
DataTransfer	Data import/export with batch processing and job queues
DataGrid	Custom data grid component with filtering, sorting, export, and saved filters
FPC	Full Page Cache with event-driven invalidation for products, categories, and CMS pages
Theme	Multi-theme support with theme customization and translation management
Notification	Order notification system with event-driven updates

5. Codebase Metrics

The Bagisto codebase is substantial, reflecting the comprehensive nature of the e-commerce framework. With over 474,000 lines of PHP code across 2,434 files, the

project represents a significant development effort. The codebase also includes 551 Blade template files for server-side rendering, 87 JavaScript files for frontend interactivity, and 31 CSS files for styling. The project does not use Vue single-file components (.vue files), instead relying on Blade templates combined with vanilla JavaScript for most of the frontend logic.

File Distribution

File Type	Count	Description
PHP Files	2,434	Core application logic, models, controllers, repositories
Blade Templates	551	Server-side views for admin panel and storefront
JavaScript Files	87	Frontend interactivity and AJAX handlers
CSS Files	31	Stylesheets for admin and shop themes
Migration Files	100+	Database schema definitions and updates
Language Files	300+	Translation files for 20+ languages

Lines of Code by Package

The largest packages by code volume are the Admin panel (553 PHP files) and the Shop storefront (329 PHP files), which together account for a significant portion of the total codebase. The Core package (171 PHP files) provides the foundational classes and interfaces that other packages depend on. The Product module is one of the most complex, supporting multiple product types including simple, configurable, grouped, bundle, downloadable, virtual, and booking products. Each product type has its own set of models, repositories, controllers, and views, contributing to the overall code volume.

Top Packages by PHP File Count

Package	PHP Files	Primary Responsibility
Admin	553	Backend administration panel with all management interfaces
Shop	329	Customer-facing storefront with catalog and checkout
Core	171	Framework foundation, base classes, and shared utilities
Product	~150	Multi-type product management system
Sales	~120	Order processing, invoicing, and refund management

Customer	~100	Customer accounts, authentication, and group management
Checkout	~90	Shopping cart and multi-step checkout flow
BookingProduct	~80	Booking system with 5 sub-types

6. Dependencies & Requirements

Bagisto has a comprehensive dependency graph that reflects its enterprise-grade feature set. The project requires PHP 8.3 or higher, which is a relatively modern requirement that ensures access to the latest language features including readonly properties, enums, and fiber support. The framework is built on Laravel 12, the latest major version of the Laravel framework, which itself introduces significant improvements in performance, developer experience, and ecosystem integration.

Server Requirements

Requirement	Minimum Version	Purpose
PHP	8.3+	Runtime environment
ext-calendar	*	Date and calendar functions
ext-curl	*	HTTP client operations
ext-intl	*	Internationalization support
ext-mbstring	*	Multibyte string handling
ext-openssl	*	Encryption and HTTPS
ext-pdo	*	Database abstraction layer
ext-pdo_mysql	*	MySQL PDO driver
ext-tokenizer	*	PHP code parsing
MySQL	8.0+	Primary database
Redis	*	Cache and session storage
Elasticsearch	7.17+	Full-text search engine
Node.js	18+	Frontend asset compilation

Key Composer Dependencies

Package	Version	Role
laravel/framework	^12.0	Core PHP framework (MVC, ORM, Queue, etc.)
laravel/sanctum	^4.0	API token authentication
laravel/socialite	^5.0	OAuth social login providers
laravel/cashier	^16.0	Stripe billing subscriptions
laravel/octane	^2.3	High-performance application server
laravel/ai	^0.2.2	AI integration for MagicAI module
elasticsearch/elasticsearch	^8.10	Elasticsearch PHP client
redis/redis	^2.2	Redis client for PHP
maatwebsite/excel	^3.1.46	Excel import/export
intervention/image	^2.4 ^3.0	Image manipulation and processing
barryvdh/laravel-dompdf	^2.0 ^3.0	PDF generation from HTML
mpdf/mpdf	^8.2	Alternative PDF generation engine
stripe/stripe-php	^17.3	Stripe payment SDK
paypal/paypal-server-sdk	^2.0	PayPal payment SDK
razorpay/razorpay	^2.9	Razorpay payment SDK
astroatomic/laravel-translatable	^11.16	Multi-language model translations
konekt/concord	^1.16	Module management system
kalnoy/nestedset	^6.0	Nested set for category trees
spatie/laravel-responsecache	^7.4	HTTP response caching
spatie/laravel-sitemap	^7.3	XML sitemap generation
khaled.alshamaa/ar-php	^6.0.0	Arabic language processing support

Development Dependencies

The development tooling includes PHPUnit 11 and Pest 3 for testing, Laravel Pint for code formatting, PHPStan via Laravel Boost for static analysis, and Laravel Debugbar for

development profiling. The project also includes Playwright for end-to-end browser testing, indicating a commitment to comprehensive testing coverage. The bagisto/laravel-datafaker package provides fake data generation for testing and development purposes.

7. Deployment Assessment

Deploying Bagisto requires careful planning due to its multiple service dependencies and the need for proper environment configuration. The framework relies on several backend services that must be properly configured and interconnected: a MySQL 8.0 database, Redis cache server, and Elasticsearch cluster. Each of these services requires specific configuration parameters and resource allocations to ensure optimal performance.

Deployment Methods

Bagisto supports multiple deployment approaches to accommodate different hosting environments and team preferences. The primary method is the traditional Composer-based installation, where developers clone the repository, install PHP dependencies via Composer, configure the .env file, run database migrations, and compile frontend assets using Vite. The project also provides a GUI installer that simplifies the initial setup process for less technical users, walking them through database configuration, admin account creation, and basic store settings.

For containerized deployments, the included Docker Compose file provides a complete development environment with all required services. The Docker setup uses Laravel Sail as the base image, which is well-documented and widely adopted in the Laravel ecosystem. For production deployments on cloud infrastructure, Bagisto offers a pre-configured Amazon Machine Image (AMI) on the AWS Marketplace, enabling rapid deployment on EC2 instances without manual setup. This AMI includes all necessary software pre-installed and pre-configured, making it ideal for teams that want to get started quickly on a scalable cloud environment.

Deployment Readiness Checklist

Component	Status	Priority
PHP 8.3+ Runtime	Required	Required

MySQL 8.0 Database	Required	Required
Redis Cache Server	Required	Required
Elasticsearch 7.17+	Required	Required
Node.js 18+ (build time)	Required	Required
Composer 2+	Required	Required
.env Configuration	Required	Required
Storage Permissions	Required	Required
SSL Certificate	Recommended	Recommended
Queue Worker (Supervisor)	Recommended	Recommended
Kibana Dashboard	Optional	Optional
Mailpit (development)	Optional	Optional

Environment Limitation

During our analysis, we attempted to run the project locally but encountered a significant limitation: PHP and Composer are not installed in the current analysis environment. Additionally, Docker is also unavailable, which prevented us from using the containerized deployment method. Node.js v24.16.0 and npm 11.13.0 are available, but these alone are insufficient to install and run the Bagisto application, which primarily requires a PHP runtime. This means that while we were able to perform a thorough static analysis of the codebase, we could not execute the application to verify runtime behavior, test functionality, or assess actual performance characteristics.

8. Security & Compliance

Security is a critical concern for any e-commerce platform that handles sensitive customer data, payment information, and business transactions. Bagisto incorporates several security measures and compliance features that address the most common security requirements for e-commerce applications. The project has a dedicated SECURITY.md file and encourages responsible disclosure of security vulnerabilities through a private reporting channel (support@bagisto.com) rather than public issue tracking.

Built-in Security Features

- **Laravel Sanctum:** Token-based API authentication with CSRF protection for SPA and mobile clients
- **Laravel Socialite:** Secure OAuth-based social login with support for major providers
- **Stevebauman/Purify:** HTML input sanitization using HTML Purifier to prevent XSS attacks
- **Enshrined/SVG-Sanitize:** SVG file sanitization to prevent XML-based attacks when uploading SVG images
- **PragmaRX/Google2FA:** Two-factor authentication support for admin accounts
- **SSL/HTTPS:** OpenSSL extension required for encrypted communications
- **Rate Limiting:** Built-in Laravel rate limiting for API endpoints and form submissions

Compliance Modules

Bagisto includes dedicated modules for regulatory compliance. The GDPR module provides data request management and consent tracking, enabling store owners to comply with the European General Data Protection Regulation. Customers can submit data access and deletion requests through a self-service interface, and administrators can manage these requests through a dedicated admin panel. The EUWithdrawal module, added in version 2.4.5, implements compliance with Directive (EU) 2023/2673 (Article 11a CRD), allowing customers and guests to withdraw from contracts online via a dedicated button on order pages. This module includes durable-medium confirmation emails, an admin datagrid with timeline view for managing withdrawals, and per-channel enable toggle. The inclusion of the khaled.alshamaa/ar-php package specifically for Arabic language processing demonstrates the project's commitment to supporting RTL (right-to-left) languages and Middle Eastern markets.

Security Considerations

While Bagisto provides a solid security foundation, there are several areas that require attention during deployment. The project uses the Laravel response cache, which can potentially serve stale content if not properly invalidated. The FPC (Full Page Cache) module mitigates this by implementing event-based cache invalidation triggered by product updates, category changes, and other content modifications. Additionally, the Elasticsearch instance in the Docker configuration has security disabled

(`xpack.security.enabled=false`), which is acceptable for development but must be properly secured for production. The MySQL configuration also allows empty passwords in the Docker setup, which should never be used in a production environment.

9. Recommendations

Based on our comprehensive analysis of the Bagisto framework, we provide the following recommendations for teams considering adopting, deploying, or contributing to the project. These recommendations are organized by priority and cover deployment, security, performance, and development best practices.

High Priority Recommendations

- **Use Docker for Development:** The Docker Compose setup provides all required services (MySQL, Redis, Elasticsearch) in a consistent environment. This eliminates the common "works on my machine" issues and significantly reduces onboarding time for new developers. Ensure Docker has at least 4GB of RAM allocated, as Elasticsearch requires substantial memory for indexing operations.
- **Secure Elasticsearch for Production:** The default Docker configuration disables Elasticsearch security (`xpack.security.enabled=false`). Before deploying to production, enable X-Pack security, configure TLS, set up authentication, and restrict network access to the Elasticsearch port. Consider using a dedicated Elasticsearch cluster with proper node sizing based on your product catalog size and search traffic.
- **Configure Queue Workers:** Bagisto uses Laravel queues for data imports, catalog rule indexing, and email sending. Configure a queue worker with Supervisor to ensure reliable background processing. Use the Redis driver for queues (already configured via Predis) for better performance compared to the database driver.
- **Set Up SSL/TLS:** E-commerce platforms must encrypt all data in transit. Obtain an SSL certificate from a trusted Certificate Authority (Let's Encrypt provides free certificates) and configure your web server to enforce HTTPS connections. The OpenSSL PHP extension is already required by Bagisto.

Medium Priority Recommendations

- **Implement Caching Strategy:** Leverage the built-in FPC module and configure Redis caching for sessions, queries, and route caching. The Spatie ResponseCache

package is already included; configure it with appropriate cache lifetimes and exclusion rules for dynamic pages like cart and checkout.

- **Set Up Monitoring:** Implement application performance monitoring using tools like Laravel Telescope (for development), New Relic, or Datadog (for production). Monitor queue worker health, Elasticsearch index status, Redis memory usage, and MySQL query performance. Set up alerts for error rate thresholds and response time degradation.
- **Review Payment Configuration:** Each payment gateway requires specific API credentials and configuration. Test all payment flows in sandbox/test mode before going live. Ensure webhook URLs are properly configured for payment confirmation callbacks. Consider implementing idempotency keys for payment requests to prevent duplicate charges.
- **Optimize Elasticsearch:** Configure custom analyzers and mappings for your product data to improve search relevance. Set up index lifecycle management to handle index growth. Consider implementing search suggestions and faceted navigation for better user experience.

Low Priority Recommendations

- **Explore Headless Commerce:** Bagisto supports headless commerce through its API, and there is an official Next.js commerce storefront (github.com/bagisto/nextjs-commerce). Consider this approach if you need a highly customized frontend or want to serve multiple client applications (web, mobile, IoT) from a single backend.
- **Leverage MagicAI for Content:** The MagicAI module supports multiple LLM providers for automated product description generation, customer support chatbots, and search enhancement. Start with OpenAI integration for product descriptions, then expand to other use cases as you gain experience with the AI capabilities.
- **Consider Multi-Vendor Marketplace:** If your business model involves multiple sellers, evaluate the official Multi Vendor Marketplace extension, which adds seller management, commission tracking, and product approval workflows to the core Bagisto platform.
- **Mobile Commerce:** Bagisto offers an open-source Flutter mobile app (github.com/bagisto/opensource-ecommerce-mobile-app) that connects to the same backend API. Evaluate this option if mobile commerce is a priority for your target market.